

BI – Sales Performance Dashboard

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Domain: Business Intelligence

Tool: Microsoft Power BI

Project Objective

The objective of this project is to develop an interactive Power BI dashboard for analyzing sales performance. The dashboard provides insights into sales, quantity, transactions, average order value, cities, product categories, store formats, and sales trends over time.

Dataset

The project uses the BrewMetrics sales dataset containing sales transaction information.

The data is organized using a **star schema** consisting of:

- Fact_Sales
- Dim_Date
- Dim_City
- Dim_Product

Data Model

The project uses a star-schema data model.

Fact table:

- Fact_Sales

Dimension tables:

- Dim_Date
- Dim_City
- Dim_Product

The dimension tables provide descriptive information that can be used to analyze the sales records stored in the fact table.

Dashboard Components

The dashboard contains:

- Total Sales
- Total Quantity

- Total Transactions
- Average Order Value
- Sales Trend Over Time
- Sales by City
- Sales by Category
- Sales by Store Format
- City filter
- Category filter
- Date hierarchy

DAX Measures

The project includes DAX measures such as:

Total Sales
Total Quantity
Total Transactions
Average Order Value
Month-over-Month Sales Growth %
Running Total Sales
City Sales Rank
Average Sales per Quantity

These measures are used to calculate and analyze different sales performance indicators.

Git and GitHub

The project was developed using the Power BI Project (.pbip) format and managed using Git.

The project was committed and pushed to GitHub using:

```
git add  
git commit  
git push
```

The repository contains the Power BI project files, dataset, README documentation, and DAX development notes.

Reflection

Working on the BrewMetrics BI project helped me understand how raw sales data can be transformed into an interactive business intelligence dashboard. I learned how to organize data using a star schema and how dimension and fact tables work together for analysis.

I also gained practical experience in creating DAX measures for total sales, quantity, transactions, average order value, running totals, month-over-month growth, city ranking, and average sales per quantity.

One of the useful parts of the project was using Copilot to generate and review DAX formulas. I learned that generated formulas should still be checked against the actual data model and requirements. For example, I reviewed edge cases such as blank or zero values before using a calculation.

I also learned how to create a date hierarchy and use it for time-based analysis. Building the dashboard helped me understand how slicers and visualizations can make business data easier to explore.

Finally, using Git and GitHub with the Power BI Project format helped me understand how BI projects can be version-controlled and documented. Overall, this project improved my understanding of Power BI, DAX, data modeling, dashboard design, and project documentation.

Challenges Faced

- Understanding the relationship between fact and dimension tables.
- Creating and applying DAX measures correctly.
- Handling blank or zero values in calculations.
- Creating and using the date hierarchy.
- Understanding how Power BI Project (.pbip) files are organized.
- Managing the project using Git and GitHub.
- Verifying that only the required project files were tracked by Git.

Learning Outcomes

After completing the project, I was able to:

- Build a Power BI dashboard.
- Understand star-schema data modeling.
- Create DAX measures.
- Perform time-based sales analysis.
- Create interactive slicers and visualizations.
- Use date hierarchies.
- Analyze sales by city, category, and store format.
- Use Git and GitHub for project version control.
- Document the development process.

Conclusion

The BrewMetrics BI project provided practical experience in converting sales data into meaningful business insights. Through Power BI, DAX, data modeling, visualization, and GitHub version control, I developed a complete and documented BI dashboard that can be used to explore sales performance from different perspectives.

